

Multiplicity: Scheffé, Tukey, Bonferroni and the False Discovery Rate

A test with error rate 0.05 keeps that promise only when it is the only inference drawn from the data. A regression analysis rarely stops at one: it compares every pair of groups, examines each of many coefficients, and follows a significant F test with the contrast that looks most interesting. This chapter measures the error of a whole family of inferences and controls it in four ways. Scheffé's method covers every estimable function in a subspace, even functions chosen after seeing the data, and is exactly the F test read one direction at a time. Tukey's studentized range is exact for all pairwise differences of balanced means. Bonferroni's inequality, and Holm's sharper step-down version, need only the level of each test. The Benjamini–Hochberg procedure controls the expected proportion of false discoveries, which is what screening many hypotheses needs. The chapter ends with a guide to choosing among them.

What you need. The normal linear model and the sampling distributions of [Chapter 7](#) (Theorem 7.5.1, Corollary 7.5.2, Corollary 7.5.3), independence of projections from [Chapter 4](#) (Theorem 4.4.4), and orthogonal projections onto subspaces from [Chapter 6](#) (Theorem 6.5.1, Theorem 6.5.2). Estimable functions and contrasts in the one-way layout come from [Chapter 8](#) (Theorem 8.7.1, Proposition 8.7.2), and the sum of squares of a linear hypothesis from [Chapter 9](#) (Theorem 9.2.4). The F test of [Chapter 11](#) (Theorem 11.1.1, Theorem 11.3.2) and the confidence intervals and ellipsoids of [Chapter 12](#) (Theorem 12.1.1, Theorem 12.2.1) are the single-inference versions of what this chapter does for families. Two pieces of machinery come in directly: Scheffé's theorem in [Section 13.2](#) is the Cauchy–Schwarz inequality together with the generalized Rayleigh quotient of [Chapter 1](#) (Proposition 1.1.1, Corollary 1.7.8), and [Section 13.4](#) uses the independence of uncorrelated normal blocks from [Chapter 3](#) (Theorem 3.4.1). The independence of orthogonal projections (Theorem 4.4.4) is named above for the joint distribution it justifies, but no proof in the chapter cites it.

13.-1.1 Roadmap

- [The multiple testing problem](#)
- [Scheffé's projection method](#)
- [Tukey's studentized range](#)
- [Bonferroni and Holm](#)
- [False discovery rate control](#)
- [Choosing a procedure](#)
- [Summary and notes](#)